SCALE FOR PROJECT RAG AGAINST THE MA (/PROJECTS/RAG-AGAINST-THE-MACHIN

You should evaluate 1 student in this team

★

Git repository

```
git@vogsphere-v2.42madrid.com:vogsphere/intra-uuid-47f57a9a-7b6
```

Introduction

- Remain polite, respectful and constructive during the evaluation.
- This project evaluates information retrieval, text chunking, answer generation and overall RAG system design.
- Reference exam scripts are provided in `exams/scripts/`: `exam_retrieval.sh`, `exam_answer.sh`, `exam_edge_cases.sh`
- See `exams/exam_scripts_instructions.md` for detailed corrector guidance.
- The correction is designed so the retrieval pipeline (~8 min) runs in the background while you inspect code and ask questions.
- Every mandatory question is pass/fail. The whole mandatory part must be validated (all questions pass) before the bonus is graded.

Guidelines

- Only grade the work present in the student's Git repository.
- Set up an evaluation workspace: clone the student repository into a subdirectory named `student/`, and place the provided `moulinette` binary and the `exams/` scripts alongside it.
- Run `cd student && uv sync` to install the student's dependencies. The student's `pyproject.toml` and `uv.lock` live at the root of their repository, so `uv sync` is expected to work from there.
- The student solution must NEVER import moulinette packages. Verify (match imports only, not the word in comments/docstrings): `grep -rnE "^[[:space:]]*(import|from)[[:space:]]+moulinette" student/ --include="*.py"`
- The program must not crash unexpectedly during evaluation.
- No modification of student code is allowed during correction.
- The datasets are provided with this scale. Students work only with the public datasets; for grading, unzip the corrector-only private datasets: `unzip data/datasets/datasets_private.zip -d data/datasets/` These private datasets are NOT expected to be in the student's repository.
- Use flags if cheating or plagiarism is detected.

Attachments

- datasets_private.zip (https://cdn.intra.42.fr/document/document/54679/datasets_private.zip)
- exams.zip (https://cdn.intra.42.fr/document/document/54680/exams.zip)
- subject.pdf (https://cdn.intra.42.fr/pdf/pdf/220267/en.subject.pdf)
- datasets_public.zip (https://cdn.intra.42.fr/document/document/54681/datasets_public.zip)
- README.md (https://cdn.intra.42.fr/document/document/54682/README.md)
- vllm-0.10.1.zip (https://cdn.intra.42.fr/document/document/54683/vllm-0.10.1.zip)
- moulinette.zip (https://cdn.intra.42.fr/document/document/54684/moulinette.zip)

Mandatory Part

Preliminaries & Setup

Check the following requirements:

- The Git repository is available and contains only the requested files
- Python 3.10+ is used
- uv is used as the project and package manager
- pyproject.toml and uv.lock files are present
- src/ directory exists with the main module
- a Makefile is present at the repository root
- README.md file is present
- uv sync completes without errors (run from the student repo root, i.e. `cd student && uv sy` evaluation workspace)
- Corrector-only step: unzip the private datasets provided with this scale: unzip data/datasets/datasets_private.zip -d data/datasets/ (the student is NOT expected to ship pri

Are all setup requirements met?

Yes No

Launch Pipeline (Background)

Launch the automated retrieval pipeline in a background terminal.
This runs indexing, search, and evaluation (~8 min total).

In a separate terminal, run:
./exams/scripts/exam_retrieval.sh \
--student-path ./student \
--moulinette-path ./moulinette-ubuntu

Continue with code inspection (Q3-Q6) while this runs.
Keep this run's output (terminal log and summary.log): questions
Q7 to Q10 read their results from it (docs/code Recall@5, indexing
time and throughput). Do not close the terminal before Q10.

Did the pipeline launch successfully?

Yes No

Code Quality & Pydantic Models

Review code quality and data model implementation. Check ALL of the following:

Code quality:

- Data models use pydantic for validation and type safety (service or orchestration classes suc indexer/retriever/pipeline need not be pydantic)
- Code adheres to flake8 coding standards, and passes mypy (run `make lint` , which the Co Instructions define as flake8 + mypy)
- The Makefile exposes the required rules (install, run, debug, clean, lint)
- Exceptions are handled gracefully with try-except blocks
- Resources are properly managed (context managers or explicit close())
- Progress bars are implemented for long-running operations (tqdm)

Pydantic models (all 8 must be correctly implemented):

- MinimalSource (file_path, first_character_index, last_character_index)
- UnansweredQuestion and AnsweredQuestion
- RagDataset
- MinimalSearchResults and MinimalAnswer
- StudentSearchResults and StudentSearchResultsAndAnswer

Output format:

- Search operations output valid JSON using StudentSearchResults model
- Answer operations output valid JSON using StudentSearchResultsAndAnswer model
- Each source includes correct file_path, first_character_index, last_character_index

Does the code meet all quality standards and are all models correctly implemented?

Yes No

Chunking Strategies

Verify chunking strategies:

- At least 2 chunking strategies are implemented (e.g., Python code + markdown/text)

- The chunk size default is 2000 characters and is configurable via --max_chunk_size
- Student can explain the impact of chunk size on retrieval performance: "What happens if chu small? Too large?"

Ask the student to show the code for both strategies.

Are chunking strategies properly implemented?

Yes No

Retrieval System

Verify retrieval implementation:

- At least one method (TF-IDF or BM25) is implemented
- System retrieves and ranks relevant information
- Student can explain how their chosen method works

Live demo: ask the student to run a search query and explain the results:
uv run python -m src search "How to configure OpenAI server?" --k 10

Is the retrieval system properly implemented?

Yes No

Answer Generation

Verify answer generation:

- Qwen/Qwen3-0.6B is used as the DEFAULT model
- Run: uv run python -m src answer "How to configure OpenAI server?" --k 10
- System passes retrieved context to LLM within token limits
- Answers are generated based on retrieved code/documentation

Note: The student may support additional local models beyond the default.
Check the README for documentation of supported models.
During evaluation, always test with the default model (Qwen3-0.6B).

Does answer generation work correctly with the default model?

Yes No

Retrieval Performance: Docs

Check the exam_retrieval.sh results.

The script evaluates Recall@5 on the private docs dataset (100 questions).
Look for the "Docs Recall@5" line in the output.

Pass criterion: Docs Recall@5 >= 80% (the threshold stated in the subject).

Does the docs dataset meet the 80% Recall@5 threshold?

Yes No

Retrieval Performance: Code

Check the exam_retrieval.sh results.

The script evaluates Recall@5 on the private code dataset (100 questions).
Look for the "Code Recall@5" line in the output.

Pass criterion: Code Recall@5 >= 50% (the threshold stated in the subject).

Does the code dataset meet the 50% Recall@5 threshold?

Yes No

Retrieval Performance: Indexing Time

Check the exam_retrieval.sh results.

Test 1/4 of the script times the full indexing run against the
300s (5 min) limit stated in the subject. Look for the "Indexing"
line in the output (or the "INDEX:" line in summary.log).

Pass criterion: indexing completed in <= 300s
(reported PASSED / "INDEX: OK", not EXCEEDED or CRASHED).

Did indexing complete within the 5-minute limit?

Yes No

Retrieval Performance: Throughput

Check the exam_retrieval.sh results.

Test 2/4 of the script times search over the two 100-question datasets (200 questions total) against the 90s limit stated in the subject. Look for the "Warm Retrieval Throughput" line (or the "THROUGHPUT:" line in summary.log).

Pass criterion: 200 questions retrieved in <= 90s (reported PASSED / "THROUGHPUT: OK", not EXCEEDED or CRASHED).

Did retrieval stay within the 90s throughput limit?

Yes No

Answer Quality

Run the answer exam script to evaluate answer quality:

./exams/scripts/exam_answer.sh \
--student-path ./student \
--moulinette-path ./moulinette-ubuntu

The script lists questions where sources were correctly retrieved, then you pick 3 questions. For each, the student's answer is displayed.

The mandatory Qwen/Qwen3-0.6B model has known reasoning limits, so grading prioritizes retrieval quality and grounding over perfect final phrasing.
A satisfactory answer should:

- be coherent and understandable,
- be mostly grounded in the retrieved sources (no major hallucination),
- address the question asked. Minor incompleteness due to base-model limitations is acceptat

Pass if at least 2 out of 3 answers are satisfactory.

Does the student pass the answer evaluation (2/3 answers ok)?

Yes No

Student Understanding

Test the student's understanding of RAG concepts and their implementation.
Ask these 5 questions (in order):

1. "What is Retrieval-Augmented Generation (RAG) and why is it useful?"
2. "Walk me through your complete RAG pipeline, from raw documents to a generated answer.'
3. "What is the difference between TF-IDF and BM25 as retrieval methods?"
4. "What implementation choices did you make and why? What trade-offs did you consider?"
5. "If you had more time, what would you improve in your system?"

Pass requires 4 out of 5 correct/satisfactory answers.
A satisfactory answer demonstrates genuine understanding, not just memorized definitions.

Does the student demonstrate good understanding (4/5)?

Yes No

README & Documentation

Verify README.md contains all required sections. Checklist:

1. [ ] System architecture (RAG pipeline components and interactions)
2. [ ] Chunking strategy explanation
3. [ ] Retrieval method details (algorithm and ranking mechanism)
4. [ ] Performance analysis (recall@k scores and system performance)
5. [ ] Design decisions and trade-offs
6. [ ] Example usage with clear instructions

If the student implemented bonus features, the README should also document those experiments and their results.

Pass if at least 5 out of 6 items are present and well-documented.

Is the README comprehensive and well-documented?

Yes No

System Reliability

Run the edge case exam script:

./exams/scripts/exam_edge_cases.sh --student-path ./student

This tests 4 degenerate inputs:

1. Empty query: search "" --k 10
2. Gibberish query: search "asdfghjkl" --k 10
3. k=0: answer "What is vLLM?" --k 0
4. Bad dataset path: search_dataset --dataset_path /nonexistent.json

Pass criterion: ALL 4 tests complete without Python tracebacks.
The program may print error messages or return empty results: that
is acceptable. Unhandled exceptions with tracebacks are NOT acceptable.

Does the system handle all edge cases without crashing?

Yes No

Bonus Part

Bonus: Advanced RAG

Graded only once the entire mandatory part is validated.

There are exactly five bonuses, each worth one point (one star). They all
run on a CPU-only campus machine. Award one star per bonus that is
implemented AND demonstrated working (not merely described in the README):

1. Semantic embeddings used for retrieval (CPU model, e.g. all-MiniLM-L6-v2)
2. Hybrid retrieval combining the lexical and semantic rankings
3. Incremental indexing (re-index only changed files)
4. Caching of the index and/or query results
5. Local HTTP API to query the index and answer questions

Star rating = number of bonuses working (0 -> 1 star, 5 -> 5 stars).
Ask the student to demonstrate each claimed bonus.

Rate it from 0 (failed) through 5 (excellent)

Ratings

Don’t forget to check the flag corresponding to the defense

✔ Ok ★ Outstanding project

Empty work Incomplete work Invalid compilation Norme Cheat

Concerning situation Forbidden function Can’t support / exp

Conclusion

Leave a comment on this evaluation (2048 chars max)

Finish evaluation

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Privacy policy (https://profile.intra.42.fr/legal/terms/5)
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